On Par: Leveraging GPR for Perfect Golf, on Average

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Abstract

This work presents a proof-of-concept framework computing player and course specific optimised golf strategies, leveraging 2D Gaussian Process Regression under various loss functions. Recognising golf as a sequential decision-making sport under uncertainty, we develop a simulation-based approach to evaluate how different combinations of club and aimpoint influence expected scoring outcomes.

Given the constraints of limited data, we describe our data generation mechanism, which estimates expected strokes to hole out by combining interpolated values from Mark Broadie's strokes gained tables with slope and tier adjustments on the green. Shot dispersion data derived from LPGA TrackMan data are overlaid onto course geometries that are digitised and transformed from satellite imagery. Our model evaluates the scoring impact of different club/aimpoint strategies, enabling strategic decisions that, under the premise of different loss functions, either minimise expected strokes or maximise birdie odds, tailored to individual players tendencies and needs, accounting for course-specific risks.

Although preliminary in scope, this project establishes a modular and scalable pipeline that can support more sophisticated modelling with larger datasets. It provides a principled foundation for a future decision support tool in golf that offers personalised, data-driven recommendations grounded in each golfer's unique profile and necessities.

1 Introduction

Golf performance is shaped by a complex interplay of biomechanical, environmental and psychological factors. While golfers are lured into the impossible mission of working towards the perfect swing (evidenced by the very many AI swing analysis projects being developed), most golfers will see more consistent improvement in their scores by tuning their strategic decisions on the course rather than perfecting their swing mechanics. From club selection to swing length to aimpoint, the sport is defined by a fine selection of nuanced choices. Not even the finest of golfers are faced with two identical shots or will hit two shots the same way – as such, the choices they make matter significantly.

This project emerges at a pivotal moment in the evolution of sports analytics. In the wake of the transformative impact of sabermetrics in baseball [Lew04], the golf world is beginning to embrace a similar statistical revolution. The development of Strokes Gained (SG) [1] by Broadie [Bro14] has redefined performance metrics in the sport. In parallel, course strategy systems like DECADE [Faw] have enabled players to make informed decisions grounded in statistical insight. The potential of statistical thinking in golf, however, is far from fully realised. As an individual sport, golf offers a perfect playground to bake in player-specific tendencies and situational risk into strategy. By exploring decision making through the Bayesian paradigm – implementing different loss functions and benefitting from the benefits of Gaussian Process Regression [WR95]

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[1] Strokes gained: Different between expected number of shots for a tour professional to hole out from a given distance and the number of shots actually taken

– this project contributes to the literature by trying to design more adaptive, intelligent, and mathematically-backed gameplay models.

Broadie's SG model revolutionised performance evaluation by enabling players to identify and address their areas of weakness by offering more interpretable and dissectable insights. Fawcett expanded on this by translating SG into actionable strategy that proved successful as exemplified by Will Zalatoris' rapid ascent from a 3000+ WAGR rank to top three in only 3 months after adopting DECADE's approach [Faw]. However, these tools are largely built on generalised estimates. What remains lacking is a tool that provides personalised recommendations, adapting strategy not just to course layouts and generalised estimates, but to the specific players' tendencies and time-specific needs (i.e. risk averse vs risk prone – playing for birdie or trying to avoid large numbers).

One notable stride in this direction is the work in Golf Strategy Optimization and the Value of Golf Skills, which developed a large-scale stochastic shortest path framework using Markov Decision Processes (MDPs) [2] to determine optimal strategies on PGA Tour courses [SG24]. Their model integrated empirical shot distributions and simulated full hole trajectories. However, their approach made the key assumption that professional players always aim directly at the pin when approaching the green – an assumption neglecting the nuance in player decision-making, such as risk aversion, aimpoint adjustments, or personal dispersion patterns.

In contrast, this work offers a player-specific decision support tool that doesn't rely on such generalisations. Instead of modelling strategy around aggregated or inferred behaviour from the professional field, this model simply builds on a lot of these averages to build the basis of the model, implementing personalised shot distributions to ultimately define strategies.

The core of our approach lies in using Gaussian Process Regression (GPR), generating a distribution of the uncertainty of expected strokes to hole out (ESHO) over the hole we're trying to strategise and evaluating dispersion patterns for different clubs and aimpoints over said distributions. Our model then identifies the decisions that will minimise ESHO at each individual point given a pin position with already recorded data and proposes the optimal club and aimpoint combinations, helping the player succeed.

2 Data

In an ideal setting, this work would have been built upon a rich tournament data set, such at that recorded by the PGA Tour using ShotLink. Simply by acquiring a golf course specific data set comprising of (x, y) coordinates for shot starting positions within close proximity of the hole (short game shots), coupled with the number of strokes each player took to hole out from said locations under specific pin positions, we would be able to fit a GPR and run our proposed model. Unfortunately, such granular shot-level data was not available for the duration of this project, as such we constructed a synthetic dataset designed to reflect realistic two dimensional scoring distributions using a hybrid approach combining course digitisation, interpolation of Mark Broadie recorded averages (2003 - 2012 PGA Tour ShotLink data), simulation and discrete inverse transformation.

Our first step was to scrape the layout of a golf course using `OpenStreetMap` and importing the data into `QGIS` for vector editing. We chose to play around with Mountain Meadows Golf Course in California due to data availability and prior knowledge of the venue. Once refined, the geometries were exported as WKT (Well-Known Text) representations, allowing us to manipulate and analyse the spatial features in `Python` using `Shapely` library and other tools. This step enabled us to classify starting lie types which would later on deem priceless.

All of the modelling was done on hole 9, a par 3, and a simulated extension of a par 4 sharing the same features around the green as the aforementioned hole (see Figure 1). This

[2] A Markov Decision Process (MDP), is a mathematical framework for modeling decision-making in situations where outcomes are partly random and decisions are made sequentially.

hole was chosen due to a rich interaction of different types of obstacles (bunkers, rough and water hazard) and the length of the hole (approximately 190 yards) which allowed for the illustration of our goals and distinct features of our model. The extension[3] of the hole was preferred over simulation on an entirely new hole, due to the fact that it gave us full liberty to add interesting features that better allowed us to compare different goal model outputs and that data for the green of the par 3 had already been generated.

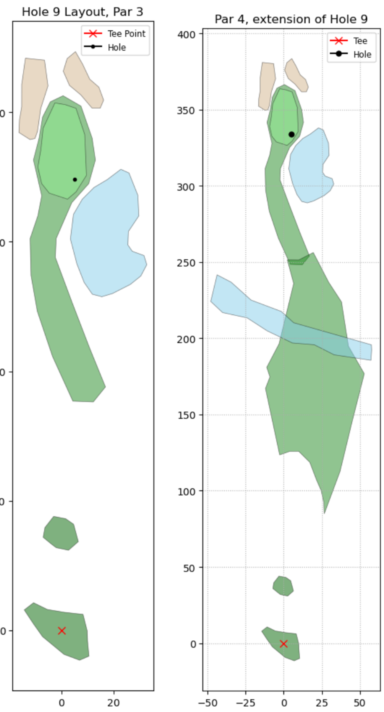

Figure 1: Par 3 assessed in our implementation, Mountain Meadows Hole 9 (left), Par 4 assessed in our implementation, extension of Mountain Meadows Hole 9 with added obstacle(right).

2.1 Green Surface, Modelling Putting

Our strategy for creating data to estimate an ESHO distribution was split into two stages aligned with our different modelling goals:

1. Minimising expected strokes to hole out
2. Maximising birdie odds

   In this case our loss function is the classic $0-1$ loss from classification, where non-birdie outcomes get penalised equally:

$$\text{Loss} = \begin{cases} 0 \text{ if birdie or better,} \\ 1 \text{ otherwise} \end{cases}$$

For both of these stages, we first constructed a synthetic but realistic putting green surface within the boundaries of the green presented above. A function was defined to generate $z(x, y)$, representing the local elevation at each point on the surface of the green (see Figure 2). The

[3]The extension involves elongating the par 3 hole to standard par 4 length (approximately 300-450 yards) while preserving the original green complex and its surrounding features. Additionally, a river obstacle was strategically placed in the middle section of the extended hole to evaluate the models' ability to detect and respond to varied hazard types and their impact on shot selection strategies, as discussed in subsequent sections.

height function was designed to include realistic features present in some tricky greens: general tilt, undulating bumps, and a mid-green tier. This was done to reflect the variability of greens, which often include multiple breaks and slopes that significantly influence the difficulty of any given putt.

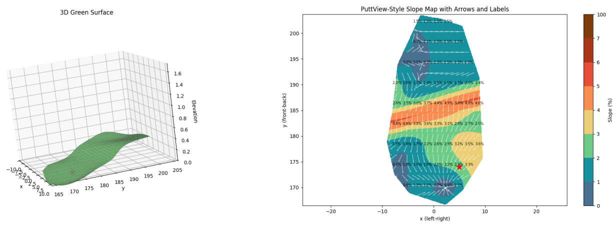

Figure 2: Two views of the putting green: 3D surface (left) and slope heatmap (right).

For the first stage of data generation, we first interpolated expected strokes to hole out as a univariate function of distance to the hole using Broadie's estimates. However, because real tournament data (which is what we would build these models on, if possible) exhibits both heteroscedasticity (increasing variance with distance) and data sparsity (fewer long putts are observed), we mimicked this by generating noisy synthetic data. This emulated how actual averages might be computed from months of tournament observations. For instance, 2-foot putts are consistently recorded with low variance, while 70-footers tend to be both rare and more variable, influenced by pin positions, slope, and course conditions.

To model the variability, we added normally distributed noise to the baseline function with a distance-dependent standard deviation. We also varied the number of samples per distance, reducing sample counts at longer ranges (not all greens offer 90 feet putts, and when they do, players do their best to hit their shots closer to the pin). The resulting simulated dataset approximated the type of empirical averages one might observe after aggregating multiple tournaments on different courses.

A Gaussian Process Regression (GPR) model was then fit to this noisy bivariate data. This allowed us to extract posterior predictive distributions of ESHO at any distance, capturing both mean and uncertainty. Once trained, this GPR model could generate posterior "slices" at any distance, forming a flexible foundation upon which course-specific adjustments could be layered.

2.1.1 Slope and Tier Adjustments on the Green

To make the GPR predictions reflect the complexity of real 3-dimensional putting surfaces, we introduced slope-based difficulty adjustments. From any ball position on the green, relative to the hole, we computed:

- **Uphill/downhill slope and severity:** by computing the slope in the direction of the line the putt would be played on;
- **Tier difference:** by classifying whether the ball and the hole were on the same tier, or whether one was higher than the other;
- **Side slope:** by drawing a line toward the hole and computing the perpendicular slope at 50% and 70% of the way along the line.

Each of these slope metrics was assigned a difficulty multiplier. For example, downhill putts with steep side slope penalised more heavily, while flat or slightly uphill putts on the same tier retained the baseline estimate. These multipliers were then applied to the posterior mean ESHO values from the GPR model. This approach effectively converted the model into a parametric framework, where the difficulty multipliers served as learnable parameters that modulated the base GPR predictions to yield an adjusted ESHO surface, more accurately reflecting the spatial difficulty of the green's landscape.

Once the adjusted ESHO distribution at a point was available, we could either report the new expected strokes directly (the posterior mean of the ESHO distribution) or simulate stroke outcomes by sampling from its posterior.

2.2 Discrete Outcome Simulation for Birdie Optimisation

The second phase of our project aimed to evaluate strategies that maximise birdie probability rather than minimise expected strokes. This required modelling the full distribution of discrete putt outcomes rather than focusing on the mean alone. To this end, we implemented a probabilistic simulation framework using distance-based one-, two-, and three-putt probabilities derived from PGA Tour averages and building on the adjusted mean, slope and tier framework outlined above.

After sampling a ball position on the green, we computed its distance to the hole and retrieved base probabilities p_1, p_2, p_3 for making the putt in 1, 2, or 3 + strokes from PGA Tour averages. We then used the previously computed adjusted ESHO for that location to shift the distribution while preserving internal consistency. This was done by redistributing probability mass from lower-stroke outcomes to higher-stroke outcomes, and sampled using a discrete inverse transformation.

This enabled us to simulate putts under realistic, course-adjusted difficulty and assign discrete scores (1, 2, or 3 putts) to sampled positions (see Figure 3). Similarly, in our birdie-optimizing model, this allowed us to compute the probability of one-putting from each position. By maximizing this probability in our shot selection strategy, we could optimize the overall likelihood of achieving birdie on the entire hole.

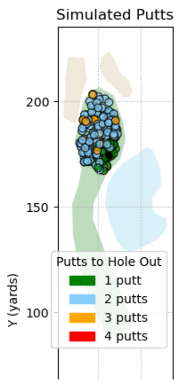

Figure 3: Putts Generated Redistributing Probability Mass

2.3 Around the Green

For shots from outside the green, where slope plays a reduced role, we reverted to a simpler modelling strategy. The ball's distance to the hole was computed, its lie type classified via polygon containment (i.e. we could identify the ball's lie given it's (x, y) location), and expected strokes to hole out were interpolated from Broadie's full-hole strokes gained tables. While less detailed, this method provides reasonable scoring estimates. In an ideal world, the GPR would simply be fit to the outcomes from different locations from historical data.

2.4 Off the Tee and From the Fairway

Details about the data used from these locations is described in Implementing GPR for Par 3s and Implementing GPR for Par 4s.

2.5 Player-Specific Data

The player-specific part of our model is derived from Trackman data from the "average" LPGA Tour player data. For the purpose of this study, in an attempt to simplify and work with the available data, we utilised stock shot data defined as the player's standard full swing carry distances for clubs ranging from a 60° wedge up to driver. Each club's dispersion distribution was modelled using approximately 30 shots recorded while aiming straight down a reference line, that is, treating the $x = 0$ axis as the player's intended target line.

Given the lack of different shot types recorded under 90 yards (players don't usually rely solely on stock wedge shots, but have clock-work systems to hit intermediate distances, i.e. if a 60° goes 70 yards, and a 54° goes 90 yards, they may have a "3/4" 54° shot practiced to hit 80 yards), we simulated approach shots under 90 yards where big distance gaps were found. We simulated plausible shot outcomes based on known shot and distance decay patterns. For example a half 60 ° wedge may fly approximately half of its full counterpart, and was estimated as roughly half the carry distance,with appropriate variance scaling (the shorter the shot, the smaller the dispersion we anticipate). These extrapolations were made conservatively in order to avoid overly optimistic shot reliability.

In an ideal setup, the player wanting to use this model would record all repeatable shot types they might call upon during play – not just full stock swings, but also fades, draws, partial swings, and speciality trajectories, that may have different preferred names. As long as the shots are executed consistently and recorded while aiming directly along the $x = 0$ axis, our framework can integrate them. The only required input is the intended target line, the starting lie, the club or shot being played and the actual end location, (x, y) of each shot.

Furthermore, we assume:

1. **Carry modelling:** We use carry distances exclusively and assume the ball stops where it lands. Roll out distances are not included, as they can vary significantly with course conditions and are difficult to model reliably without detailed surface data.

2. **No obstacle modelling and shot trajectory modelling:** Trees and other obstructions are not included in this version of the model. This allows any shot to be played over any area, regardless of vertical clearance. Along the same line, shot trajectory and apex height are ignored. Thus, the model is not yet capable of accounting for height requirements, such as clearing trees or elevated bunkers. This could be implemented in a further extension of this work.

3. **No curvature modelling:** We do not account for the shape or curvature of the ball's flight. Instead, we focus on where the ball ends up relative to the target line. Curved shots (e.g., fades, draws) are handled only if explicitly labelled and recorded as separate stock shot types.

4. **Wind, Altitude, etc :** Wind, height above sea level and other competing factors are also ignored.

In this preliminary model, shots from rough, deep rough, or sand, as in the case of intermediate shots, were not fully represented in the TrackMan data. To approximate player behaviour from the rough, we took the corresponding fairway dispersion pattern and inflated its variance while modestly reducing average carry distance.

While our model did not incorporate shot data from deep rough or sand – due to lack of access to reliable TrackMan data from these lies – this limitation does not meaningfully constrain the scope of the framework. The underlying modelling pipeline is fully generalisable to any shot type or lie condition, provided the player has recorded a sufficient number of outcomes under those contexts. As long as shot data is recorded consistently with reference to a known target line, the same dispersion modelling and simulation logic apply without modification.

It is also worth noting that the player-specific data used here could, in principle, be replaced or supplemented with shot-level data from the PGA Tour's ShotLink database or similar systems. This would offer the added benefit of capturing shot patterns under competitive, high pressure conditions – arguably providing even more relevant behavioural data than those gathered during training or practice sessions. The flexibility of the framework allows for any player's recorded shot patterns – whether simulated, tracked on a launch monitor, or observed during tournament play – to serve as input for personalised, data-driven strategic decision-making.

3 Implementing GPR for Optimal Par 3s

Our strategic optimization process for par 3 holes in this model follows a green to tee philosophy, whereby, knowing what the optimal choices near the hole are can inform the choices to be made further away from the hole. In other words, by identifying "bad" and "good" areas around the hole, we can better understand what areas we should try to land our approach shots in.

Once on the green, the strategy is straightforward: putt the ball as close to the hole as possible and leave your ball in spots where you would be comfortable putting next. The GPR distribution that we will describe next can help inform what areas are preferable. Around the green, we simplify by assuming the player executes their best available short game option, which is typically a matter of player style and is difficult to record and therefore model precisely given the variability in technique and context.

From here, we start working on our approach modelling.

At the core of this strategy lies our simulation based evaluation of shot outcomes using Gaussian Process Regression (GPR). Readers unfamiliar with this sort of regression can refer to Appendix A for a conceptual overview and explanation of prior, posterior and predictive distributions as well as the benefits of its implementation here.

3.1 Fitting GPR to Green Data

We begin by fitting a 2D GPR model to the generated ESHO data on the green. The model is trained using the `GPyTorch` library [**GPyTorch**] with a Radial Basis Function (RBF) kernel and zero mean prior. We allow the model to optimise hyperparameters (such as length scale and output scale) via marginal log-likelihood maximisation using gradient-based methods. This model provides a smooth, continuous estimate of expected strokes to hole out across the green and can be evaluated at any (x, y) location within its support (see Figure 4).

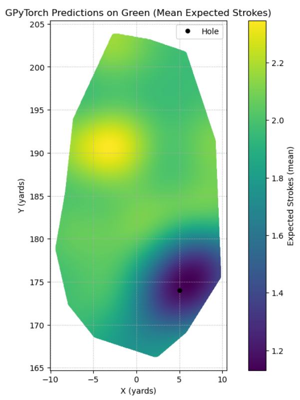

Figure 4: GPR Distribution for Expected Putts to Hole Out Generated with Generated Data

When a simulated shot lands on the green, its expected value is directly computed by looking at the posterior mean surface of the GPR model for our green. Note that each one of these green GPR model is dependant on the pin position.

3.2 Evaluating Shots Outside the Green

For balls landing outside of the green, we evaluate expected outcomes in the following way:

1. **Lie identification:** We classify the ball's lie (fairway, bunker, rough, etc.) by checking which one of the WKT pre-labelled course polygons contain the coordinate the ball lands on.

2. **Distance interpolation:** Given lie and location, we compute the Euclidean distance from the ball to the hole and interpolate the expected strokes from Broadie's published values for that lie.

3. **Water hazard:** If the ball lands in a water hazard, we draw a straight line from the ball back to the tee and determine the closest intersection with the water's boundary. We assume the player takes a drop as close as possible to this point and plays from there. As such we add a penalty stroke, reclassify the lie, and evaluate as outlined above, from this new location.

3.3 Sampling Player-Specific Shot Outcomes and Aimpoints

To simulate potential outcomes from the tee, we rely on club-specific and therefore player-specific dispersion patterns. Prior to all simulation, we compute a bivariate Gaussian distributions over landing points for each club, defined by empirical mean and covariance matrix of recorded shot data (see Figure 5).

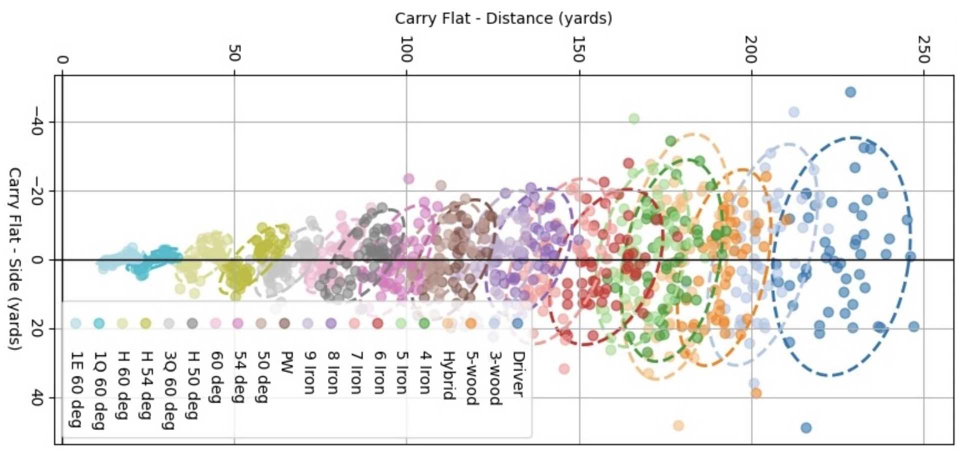

Figure 5: LPGA Tour player shot dispersion by club rotate bu 90 degrees. Each dot represents a single shot's landing point in carry (x) and side (y) coordinates. Covariance ellipses summarize the typical dispersion pattern for each club, revealing directional accuracy and distance variability. Longer clubs generally exhibit greater dispersion, while shorter clubs produce tighter shot clusters.

For a given par 3, we select the five clubs whose average carry distance most closely matches the Euclidean distance from the tee to the hole. These are the clubs most likely to be considered by the player.

For each club:

1. We simulate 100–1000 shots by sampling from the corresponding bivariate Gaussian distribution.

2. We iterate over a range of aimpoints by rotating the shot pattern between $-20°$ and $+20°$ relative to the straight line to the hole (i.e. aiming straight at the pin would be $0°$). For

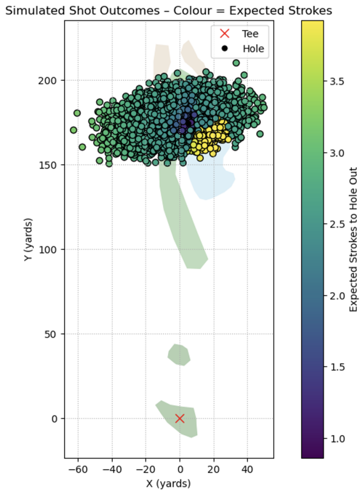

Figure 6: Example overlaying shot distribution of 5-iron aimed 2 yards left of the pin on our Par 3, evaluating expected shots to hole out at each landing point.

interpretability, the aimpoint for each of these shots is saved, not in degrees, but rather as the yards aiming due left or right of the pin. This calculation is made using standard trigonometric conversions.

3. For each angle, we apply a rotation matrix to shift the shot distribution accordingly.

4. Each shot landing location is then evaluated using the process described in Steps 1 and 2 to compute its ESHO.

3.4 Choosing the Optimal Strategy

For each club/aimpoint combination, we compute the average ESHO and add 1 to account for the shot taken from the tee. Among all combinations, we identify:

- The overall optimal club and aimpoint that minimises ESHO;
- The optimal aimpoint for each individual club, to support player preferences or constraints.

Put in mathematical terms, if we let $S_{x,y} = \mathbb{E}(\text{Strokes to hole out from } (x,y))$ and $\omega_{c,\theta,i} = (x,y)$, the point on which shot number i, hit with club c, aimed at θ (relative to the pin) lands. Then we are trying to find:

$$Opt(x,y) = \arg\min_{c,\theta} \left(\frac{1}{M} \sum_{i=1}^{m} S(\omega_{c,\theta,i}) \right)$$

This process provides a complete and interpretable club/aimpoint recommendation tailored to the player's dispersion characteristics and the course geometry. The simulation engine can be run iteratively or interactively to explore trade-offs between aggressiveness, conservatism, and shot confidence.

4 Implementing GPR for Optimal Par 4s

Perhaps more innovative than the implementation of GPR on Par 3s, is that of longer holes that require chained decision making. Our optimisation strategy for par 4 holes is simply an extension to the logic developed in our par 3 setting, continuing the process of working backward – from the green, to approach shots, to eventually selecting the optimal tee shot.

The key idea is to precompute the optimal strategy from every plausible landing position on the fairway or surrounding area, and then use this information to decide which choice from the tee will offer the best expectation for our score.

4.1 Precomputing Optimal Approaches

To begin, we evaluate a dense grid of possible second shot locations, ranging from 70 to 320 yards from the tee in 5 yard increments both "laterally" and "vertically". Each of these points is treated as a "temporary tee box" from which we simulate approach shots to the green, following the exact procedure described for par 3 holes. That is, at each location:

- We determine the lie type (fairway, rough, sand, etc.) and apply the corresponding shot dispersion model for each club.
- We simulate multiple shots (100-1000) from each club, rotating the distributions to test multiple aimpoints, and evaluate expected strokes to hole out (ESHO) based on GPR for green regions or Broadie interpolation elsewhere.
- We record the optimal club/aimpoint pair for each location: the one that results in the lowest ESHO, accounting for the one shot it takes to arrive at that point.

If our starting grid point finds itself in the water, we again compute the closest intersection to the ball from the line drawn from the tee, add a penalty stroke and reassign the drop location accordingly by evaluating the optimal club/aimpoint combination from there.

This grid gives us a complete map of optimal average outcomes from anywhere a tee shot might reasonably land. Unlike par 3s, where all shots originate from a known fairway tee box, par 4s require evaluating approach strategies from varied lies and terrain features – making this an essential part of our model.

4.2 Fitting a GPR to the Strategy Surface

Once we have computed the optimal strategy (club and aimpoint pair with corresponding ESHO) from each location on our grid, we fit a second GPR model to this surface. In this case, the model inputs are the (x, y) coordinates of the shot landing position, and the outputs are the corresponding optimal expected strokes to hole out from that location.

This fitted GPR model provides a continuous and smooth surface (a "strategic blanket") describing the optimised ESHO achievable from anywhere on the hole (see Figure 7). The kernel is again chosen as a radial basis function (RBF), and we allow the model to optimise hyper-parameters via marginal log-likelihood maximisation using gradient-based methods starting off from a 5 yard lengthscale guess to ensure sensitivity to small but meaningful differences in club selection. For example, the optimal strategy from 70 vs. 71 yards is often very similar, but a difference between 70 and 75 yards often corresponds to choosing a different shot.

4.3 Evaluating Tee Shot Options

With the GPR-fitted strategy surface in place, we can now simulate a wide range of possible tee shots, spanning from long clubs (e.g., driver, 3 wood) to more conservative options (e.g., hybrid, long iron). For each club:

1. We simulate 100–1000 tee shots for each club and aimpoint in a similar fashion to before.
2. Each shot landing location is evaluated using the GPR strategy surface described above, yielding the expected strokes from that point onward.

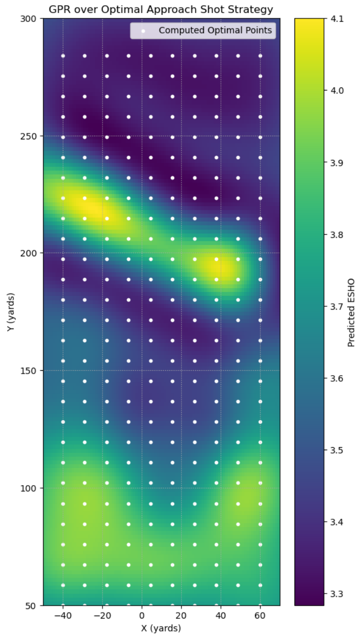

Figure 7: ESHO GPR built on optimal approach choice computed at white points

3. We add 1 stroke to account for the tee shot itself.

Finally, we compute the average ESHO for each possible club/aimpoint combination and identify the option that minimises this value. In practice, this results in recommendations such as "hybrid aimed 10 yards left of the pin" or "driver aimed straight at the pin," (maintaining the pin location as a reference we hope the player will always be able to observe) depending on the simulated risk/reward tradeoff encoded in the player's dispersion patterns and the hole's architecture.

In mathematical terms:

$$Opt_{Par4Tee} = \arg\min_{c,\theta} \left(\frac{1}{M} \sum_{i=1}^{M} Opt(x_i, y_i) \right)$$

4.4 Playing the Par 4

Our model generates a strategic roadmap for the entire par 4. From the tee, the player is given a club and aimpoint recommendation for the tee shot based on the simulations that account for the player's dispersion and the value of each possible landing area.

Next, recognising golf as a game of imperfection, we do not assume the player will always land the ball exactly where planned. Instead, the strength of this model lies in its flexibility: no matter were the ball rests after it's struck, the player can consult the optimal strategy data that the GPR surface was originally built on. Finding the closest point, the player can retrieve the optimal club and aimpoint recommendation for that specific location (see example in Figure 8).

The paradigm does not enforce a fixed sequence of instructions, but rather a dynamic and adaptive strategy that adjusts in real time based on actual shot outcomes. In such a way the player is always operating from the best known information, with access to data driven guidance on every shot all the way to the hole.

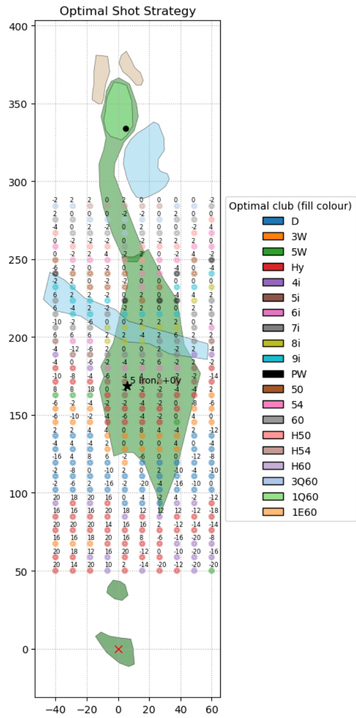

Figure 8: Optimal club choice (dot colour) and aimpoint (number above dots) for fairway shots, and recommended tee shot, 5 iron aimed straight at pin (right)

5 Loss functions: from Minimising Expectation to Maximising Birdie Probability

Up until now, our optimisation framework has focused on strategies that minimise the expected number of strokes to hole out from a given location. This mechanism offers strategies that help players strike the balance between being too aggressive and potentially blowing up a hole, and being too risk-averse, missing out on potential scoring opportunities. This framework is well suited for players seeking consistent, long term efficiency – i.e. the best score on average. However, golf is not necessarily always played under such neutral conditions. There are moments, where carding a par is not enough.

Imagine a scenario in which a player needs to make birdie on the final hole to: make the cut; force a playoff; or to outright win a tournament. In these moments, expected value is simply irrelevant. What matters is the odds of converting that single birdie opportunity which could result in clinching a win. The downside - bogey, double, or worse – is inconsequential if anything above birdie means losing. The loss function must reflect this urgency, and the model must take this into consideration.

In response to these scenarios, we reformulate our model to maximise the odds of birdieing the par 4. Redefining success as birdie and nothing else.

5.1 Birdie Probability on the Green

On longer par 4s, the only way to score birdie or better is to, at most, have one putt. As such, our effort to compute the strategy which maximises our birdie odds, start off once again, from the green. We begin by using our simulated green data to fit the probability of holing a putt on the green, $\mathbb{P}(\text{1-putt}|(x, y))$. Each point of data is labelled 1 if the putt was holed, and 0 otherwise. A Gaussian Process Regression is then fit to this binary outcome, allowing us to query any location on the green for its associated one putt probability (see Figure 9).

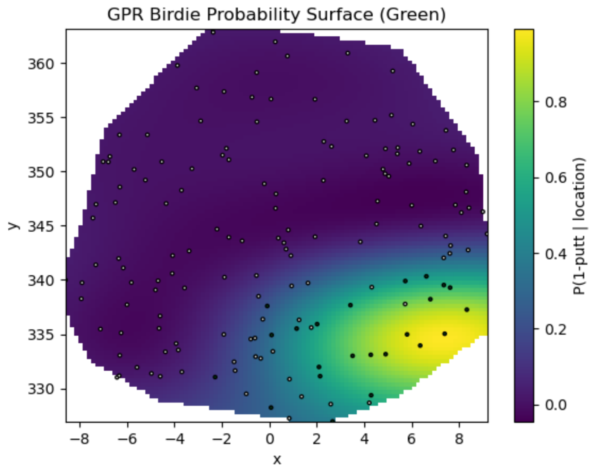

Figure 9: GPR on Birdie Odds Given Putt Landing Spot.

If a shot lands on the green, we can return this posterior mean as its birdie potential. If a shot misses the green entirely, we assign the likely birdie odd of zero. This reflects a simplifying assumption: shots that result in the highest birdie odds are unlikely to come to rest off the green or far from the hole.

5.2 Birdie Focused Approach Simulation

For each potential approach shot location, as in the case of minimising expectation for par 4s,

1. We simulate the possible shots from a selection of clubs and aim offsets.
2. Each shot is evaluated: if it lands on the green, we extract $\mathbb{P}(\text{1-putt}|(x, y))$; if not, the birdie probability is assumed to be zero.
3. The average birdie probability across all simulations is computed.
4. We retain the club/aimpoint pair that maximises this value.

As in the previous section, this process is repeated over a dense grid of locations, generating a full map of the maximum birdie probability location from each point.

5.3 A Familiar Method to Choose the Birdie Maximising Tee Shot

We then fit a second GPR model to this optimised map, learning a continuous surface over (x, y) coordinates that returns the best birdie probability from that location onward. The birdie surface, allows us to project birdie potential from all tee shot landing points, even before the player has taken the shot.

Finally, we simulate tee shots from every relevant club in the player's bag. For each:

1. We sample the dispersion pattern of the tee shot over a range of aimpoints.
2. Each landing point is queried against the birdie surface to get the "downstream" birdie odds.
3. We average the results across all simulations.

The club and aimpoint combination that maximises expected birdie probability (regardless of risk) is returned as the recommended tee shot (example of outcome, Figure 10).

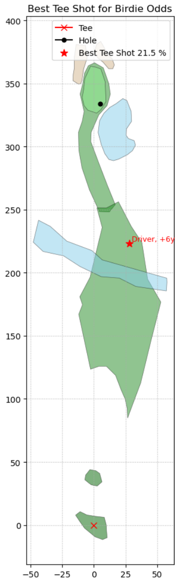

Figure 10: Optimal Tee Shot Strategy, to maximise birdie odds. As can be observed, the strategy is a lot more aggressive: choosing driver over 5 iron. This strategy's posterior mean indicates an average of over 20 % chance of ending up in birdie, assuming that the player then plays optimally from whatever the landing location ends up being (as in previous cases, the other optimal choices can be displayed in these plots)

6 Conclusion

This project set out to explore whether a simulation-based, player-specific, and loss-aware model could meaningfully optimise golf strategy across a hole in an interpretable way. By using Gaussian Process Regression (GPR) to generate smooth surfaces encoding expected strokes to hole out or the odds of securing a birdie or better, and simulating player specific dispersion patterns over digitised course layouts, we built a pipeline that not only recommends optimal club and aimpoint combinations, but does so in a way that adapts to individual tendencies, course geometry and innovatingly, scoring objectives.

Beginning in a simple green surface regression, the model expanded into a full recursive framework. The ability to work from hole to tee – constructing optimal strategies at each point and layering that knowledge to inform upstream decisions – has proven to be a powerful paradigm. The strategy proposed through this approach transforms into something dynamic: no longer a fixed set of heuristics, such as hitting it as far a possible, but a function of both where the player is and what they want to achieve. Whether minimising average strokes or maximising birdie odds, the framework can reorient itself to reflect player's intent – offering conservative plays when playing most generally, or aggressive lines when chasing necessary birdies.

Despite its contributions, the model remains preliminary in scope. It assumes clear airspace: there are no trees to shape around; no out of boundary limits; no need to clear bunkers with height, and no consideration for wind, temperature, or altitude – all of which heavily influence shot behaviour in on golf courses. Likewise, we do not currently simulate rollout or recovery shots, meaning that some high risk plays (e.g., left of the fairway, potentially into a forest) may be underestimated in cost. The model also assumes that shots behave as bivariate distributions of carry distances only; trajectory and curvature are not yet part of the decision logic. These are not trivial omissions – but they are also modular to add. With additional data, and some additions, the same framework could incorporate these complexities.

Perhaps more importantly, the model has yet to be tested in the real world. While dispersion patterns were drawn from reasonable approximations of LPGA Tour players, the pipeline has not yet been connected to ShotLink-level tournament data. Nor has it been validated on course with actual players. Doing so would offer a richer understanding of both performance and usability: would players trust a model's recommended aimpoint? Would they learn from it? Would they override it? These are open questions that move beyond code and into human behaviour.

Still, this work lays the foundation for a new kind of strategy engine in golf. It offers a principled alternative to anecdote, guesswork and experience – one that builds on the same logic as strokes gained, but does so with personal relevance. From par 3s to longer par 4s, from safe plays to must-score situations, this framework shows that meaningful strategy can be simulated, visualised, and made actionable.

There remains so much to explore: how to build in going for it decisions on par 5s; the physical consequences shifting shot distributions when not hitting balls off of flat lies; how to model recovery logic; and even how to tune a player's strategy over time. But as a proof of concept, this work affirms the possibility for software to provide actionable insights to players with reduced experience, or enhanced suggestions to even the most seasoned golfers. In a sport where every shot counts, and margins are razor-thin, building a system like this, could change it all.

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Appendix A

On Gaussian Process Regression

When performing standard linear regression, the statistician is trying to perform inference on the parameters β which, when applied to a given set of features, best describe the relationships present in the data. While computationally efficient, linear regression has several limitations: it assumes a specific functional form (linear in the parameters), requires us to choose features a priori, and, as the number of these features increases, our interpretation of the given parameters drastically decreases.

Gaussian Process Regression, takes a fundamentally different approach, focusing on making predictions. Instead of performing inference on a finite parameter space ($\beta \in \mathbb{R}^{p+1}$), GPR recognises that we primarily need predictions at specific test points, X^*, within our support.

As such, GPR doesn't model the full data generating process, instead we model a distribution of functions at the given test points (X^*), that could potentially explain our data. It is valuable to note that I could get to the same place as linear regression: if I know

$$\mathbb{E}(Y|X)$$

at all X, then I can figure out β.

We realise that by placing a distribution on β, we can induce a distribution of functions that can explain our data. Say we define our model:

$$f = \phi(x)^T \beta$$

Where $\phi(x)$, is our input vector which could take any form, e.g. $\phi(x) = \begin{bmatrix} x \end{bmatrix}$, $\phi(x) = \begin{bmatrix} x \\ x^2 \end{bmatrix}$ or

$\phi(x) = \begin{bmatrix} \sin(x) \\ \sqrt{x} \\ \vdots \\ x \end{bmatrix}$. In this case, for sake of illustration, we will let $\phi(x) = \begin{bmatrix} \phi_1(x) \\ \phi_2(x) \end{bmatrix}$. Similarly, in this case, $\beta = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$.

Now, if we put a prior on β, that is, in this case we assume:

$$\beta \sim \mathcal{N}(\vec{0}, \sigma^2 I)$$

Then we can induce the prior on $f(x) = \phi(x)^T \beta$ by computing its mean and covariance structure, given that it's normal because β is normal. By linearity of expectation, we have that:

$$\begin{aligned} \mathbb{E}[\phi(x)_i^T \beta] = 0 \;\forall\, i \implies \mathbb{E}[\phi(x)^T \beta] &= \phi(x)^T \mathbb{E}(\beta) \\ &= \phi(x)^T \cdot \vec{0} \\ &= \vec{0} \end{aligned}$$

which shows that our induced prior has mean 0.

Given our example, the assumed structure of our function is:

$$\begin{aligned} f(x) &= \phi(x_i)^T \beta \\ &= \begin{bmatrix} \phi_1(x_i) \phi_2(x_i) \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} \\ &= \phi_1(x_i)\beta_1 + \phi_2(x_i)\beta_2 \end{aligned}$$

Say we want to compute the following covariance, $\mathrm{Cov}(f(x_p), f(x_q))$, we compute the following:

$$\mathrm{Cov}(f(x_p), f(x_q)) = \mathrm{Cov}(\phi_1(x_p)\beta_1 + \phi_2(x_p)\beta_2, \phi_1(x_q)\beta_1 + \phi_2(x_q)\beta_2)$$

Since $\mathrm{Var}(\beta) = \sigma^2 I$, unless the βs match, the values are 0, therefore

$$\begin{aligned} \mathrm{Cov}(f(x_p), f(x_q)) &= \phi_1(x_p)\phi_1(x_q)\mathrm{Var}(\beta_1) + \phi_2(x_p)\phi_2(x_q)\mathrm{Var}(\beta_2) \\ &= \phi_1(x_p)\phi_1(x_q)\sigma^2 + \phi_2(x_p)\phi_2(x_q)\sigma^2 \end{aligned}$$

We notice that this is the dot product of the features! Choosing $\mathrm{Var}(\beta) = \frac{\sigma^2}{N} I$ to prevent ever growing variance, we get:

$$\mathrm{Cov}(f(x_p), f(x_q)) = \frac{\sigma^2}{2} \phi(x_p) \cdot \phi(x_q)$$

Having outlined how inducing a prior on β, may help us induce a prior on the functions that can describe our data we continue to explore the rest of this regression process.

Implementing GPR using a radial basis function kernel, requires that we consider infinitely many smooth functions that could describe our data, as such, the feature space we consider must allow for such flexibility. Instead of considering two feature vectors, which consider a feature space consisting of a collection of normal looking distributions centered at c_i with variance dictating their spread. Our feature space will consist of features in the form of:

$$\phi_t(x_i) \propto \exp\left(-\frac{x_i - c}{2l^2} \right)$$

Choosing an appropriate lengthscale l, with a combination of these functions, we can make formulate any "smooth function", given that we remember to scale prior variance (choose variance appropriately, as above).

At some point, to consider any smooth function, however, we will have to compute:

$$\mathrm{Cov}(\phi(x_q), \phi(x_p)) = k(\phi(x_q), \phi(x_p)) = \frac{\sigma^2}{N}\phi(x_p) \cdot \phi(x_q) = \frac{\sigma^2}{N}\sum_{i=1}^{n} \phi_i(x_p)\phi_i(x_q)$$

Which for very large values of N, can become ridiculously computationally expensive. As such we recur to something called the kernel trick. If we normalise the support to be $(0, 1)$ and we partition this finely enough, we can think about it in the way we would think of a Reimann sum, remembering to assign to each of these basis functions a prior variance of $\frac{\sigma^2}{N}$ due to the fact that we have N of them.

$$k(x_p, x_q) = \frac{\sigma^2}{N}\phi(x_p) \cdot \phi(x_q) = \frac{\sigma^2}{N}\sum_{i=1}^{N} \phi_{c_i}(x_p)\phi_{c_i}(x_q)$$

$$\to \sigma^2 \int_0^1 \phi_c(x_p)\phi_c(x_q)dc = \sigma^2 \int_0^1 e^{-\frac{(x_p-c)^2}{2l^2}} e^{-\frac{(x_q-c)^2}{2l^2}} dc = \sqrt{\pi} l \sigma^2 e^{-\frac{|x_p-x_q|^2}{4l^2}}$$

We have reduced costly multiplications into simple algebra, enabling us to compute the covariance structure of infinitely flexible and smooth functions with simple calculus and what we will now call a kernel function.

$$k(x_p, x_q) = \sigma^2 e^{-\frac{|x_p-x_q|^2}{l^2}}$$

This kernel function has a particular name, Radial Basis Function. It allows us to evaluate an infinite-dimensional feature space consistsing of basis functions of normal densities centred at all possible in our support. The lengthscale, l, is a way of communicating how flexible we believe our data/function to be. By having smaller values, correlations go to 0, allowing these functions to exhibit "wilder" or more jagged behaviour, and also a larger confidence bound in our fit (it could go anywhere!). As l goes to infinity, however, the functions we fit become smoother, fitting closer to lines as correlations go to $e^0 = 1$. As such, this function evaluates two data points, or xs in our support and projects how closely correlated their outputs are going to be, with small length scales, two different values close together can produce wildely different outputs and conversely.

Once we have known pair values $(x_1, y_1), (x_2, y_2) \ldots, (x_{47}, y_{47})$ for example, and I want to know how the GPR evaluates $x_{48}, \ldots, x_{470}$, I can build the function space posterior:

$$\begin{bmatrix} f(x_1) \\ \vdots \\ f(x_{47}) \\ f(x_{48}) \\ \vdots \\ f(x_{470}) \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} y_1 \\ \vdots \\ y_{47} \\ \mathbf{0} \end{bmatrix}, \begin{bmatrix} \eta & & & \\ & \ddots & & K_{12} \\ & & \eta & \\ & K_{21} & & K_{22} \end{bmatrix} \right)$$

$$= \mathcal{N}\left(\begin{bmatrix} y_1 \\ \vdots \\ y_{47} \\ \mathbf{0} \end{bmatrix}, \left[\begin{array}{cccc|c} k(x_1,x_1)+\eta & k(x_1,x_2) & \cdots & k(x_1,x_{47}) & \\ k(x_2,x_1) & k(x_2,x_2)+\eta & \cdots & k(x_2,x_{47}) & \\ \vdots & \vdots & \ddots & \vdots & K_{12} \\ k(x_{47},x_1) & k(x_{47},x_2) & \cdots & k(x_{47},x_{47})+\eta & \\ \hline & & K_{21} & & K_{22} \end{array}\right] \right)$$

Where K_{12} is the observed covariance between observed and test points, K_{21} is the observed covariance between test points and observed, and K_{22} is covariance among test points. η, is the observed variance.

Having constructed this multivariate normal density, we can exploit the known conditional distribution, which is the vector-valued version of a standard probability result, allowing us to compute the posterior predictions.

Why GPR works here!

Gaussian Process Regression is a tool that works intuitively with this project because golf courses are inherently spatial environments where nearby locations tend to have similar strategic values. GPR's smoothness assumptions align perfectly with this truth – in most cases, spots one yard apart should have similar expected strokes to hole out value. This spatial coherence is exactly what a well adjusted kernel function can capture, providing smooth interpolation across different lies. Along a similar line, the fact that Gaussian Process Regression can distinguish between high-confidence predictions (where data is dense) and uncertain regions through its posterior variance can be crucial for golf, where uncertainty can directly relate to risk.

Golf scoring relationships are complex and non linear. A putt from 10 feet isn't necessarily twice as hard as one from 5 feet. Similarly, two 100 yard shots don't necessarily have the same difficulty. Complexity doesn't scale linearly, and as such GPR can learn these arbitrary functional relationships without assuming a parametric form, truly having the capacity to adapt to the true complexity of 3-dimensional golf course surfaces.

Alternative approaches would be too rigid, or demand too much data, failing to provide the uncertainty estimates for risk aware golf strategy. GPR gives us smooth probabilistic surfaces, capturing the spatial nature of golf all while being malleable to our different optimisation mechanisms.

Glossary of Golf Terms

Lie: The ground condition or surface where a ball rests. Common lies include:

Fairway: closely mown grass.

Rough: longer grass bordering fairway lies, thicker and more unpredictable.

Sand/bunker: obstacle or hazard filled with sand.

Green: short grass area around the hole used for putting, that is, rolling the ball into the hole.

Club: instrument used to hit golf balls. Clubs vary in loft and length, affecting distance and shot shape. Examples used in this paper include: 9-iron, 8-iron, and 7-iron.

Tee/Tee box: Designated starting area for each hole.

Pin/Flag: The flagstick marking the hole's location on the green.

Hazard: Course obstacles like water or sand bunkers that penalise errant shots.

Drop: Placing the ball back in play after it lands in a water hazard, typically incurring a penalty stroke.

Par: The expected number of strokes a skilled golfer should take to complete a hole. A par-3 hole expects a golfer to finish in 3 shots.

Birdie: The score that is one stroke under par on a hole. For example, if a hole has a par of 4, and a player completes it in 3 strokes, that's a birdie.

Bogey: The score that is one stroke over par on a hole. For example, if a hole has a par of 4, and a player completes it in 5 strokes, that's a bogey.

Stroke: A single swing attempt at the ball.

Stock shot: A player's standard, repeatable "full swing" given a club.

Approach shot: A shot intended to reach the green, or "approach" the pin.

Carry distance: The distance a ball travels through the air before the it first touches the ground.

Dispersion: The spread of a player's shots around their intended target, variable depending on golf club difficulty and individual golfer's tendencies.

Aimpoint: Directional target chosen by the player, defining the centre of their shot dispersion.

Short game: Shots played close to the green (chipping, pitching, putting)

Break: The curve a putt takes due to green slope.

Trackman: A golf radar system that captures detailed shot data, including launch angle, ball speed, spin, dispersion and more.

ESHO: The number of strokes a player is expected to take from a given location until the ball is in the hole.

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